

ASP.NET Core

MSSQL Authentication Mode & Connection String

Introduction

- The Connection string is used to connect to the database
- In this tutorial we will use SQL Server, Visual Studio, C#
- We will show how we use Connection String in an ASP.NET Core Web Application

SQL Server

SQL Server

- SQL Server is a Database System from Microsoft. SQL Server comes in different editions, for basic, personal use
- SQL Server Express is recommended because it is simple to use, and it is free.
- Latest version is SQL Server 2019.

SQL Server Installation

- During the setup of SQL Server you should select "**Mixed Mode**" (i.e., both "SQL Server Authentication" and "Windows Authentication") and enter the password for your sa user.
- "Windows Authentication" is the default option during installation, so make sure to "Mixed Mode" (i.e., both "SQL Server Authentication" and "Windows Authentication") and enter the password for your sa user
- **Make sure to remember the sa password!**

SQL Server Installation - Mixed Mode

- During Installation of SQL Server: Select “**Mixed Mode**” (i.e., both SQL Server Authentication and Windows Authentication)
- Make sure to remember the “**sa**” Password!
- “sa” is short for **S**ystem **A**dministrator

SQL Server Installation - Mixed Mode

SQL Server 2019 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

Install Rules
Feature Selection
Feature Rules
Instance Configuration
Java Install Location
Server Configuration
Database Engine Configuration
Consent to install Microsoft R...
Consent to install Python
Feature Configuration Rules
Installation Progress
Complete

Server Configuration | Data Directories | TempDB | Memory | User Instances | FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

Enter password:

Confirm password:

Specify SQL Server administrators

XPS12-9Q33-LT\XPSUser (XPSUser)

SQL Server administrators have unrestricted access to the Database Engine.

< Back Next > Cancel

Annotations:

- Chỉ xác thực và đăng nhập vào SQL Server bằng tài khoản login Windows
- Cho phép xác thực và đăng nhập vào SQL Server bằng tài khoản login Windows và tài khoản sa
- Nên nhớ hay ghi lại password : ví dụ Sql2019
- Nếu không thấy tài khoản login Windows thì click

Authentication

Visual Studio

- In **WinForm** Desktop Applications you should put the Connection String in the **App.config** file
- While for **ASP.NET Core** Web Applications the Connection String should be placed in the in the **appSettings.json** file.

Authentication Methods

SQL offer 2 different Authentication methods:

- SQL Server Authentication
- Windows Authentication

SQL Server Authentication

Connection String - SQL Server Authentication

Using "SQL Server Authentication" the Connection String looks like this:

```
DATA SOURCE=<SQL Server Name>;DATABASE=<Database Name>;UID=sa;PWD=<Your Password>;
```

Replace <SQL Server Name> with the name of your SQL Server, typically "<YourComputerName>\SQLEXPRESS" if you are using SQL Server Express.

UID is a SQL Server user, here you can create your own SQL Server user inside SQL Server Management Studio or use the built-in sa user (sa=System Administrator). During the setup of SQL Server you need to select "Mixed Mode" and enter the password for your sa user.

It may look something like this:

```
DATA SOURCE=DELLPCWORK\\SQLEXPRESS;DATABASE=MEASUREMENTS;UID=sa;PWD=Password123;
```

Localhost

If you don't know the name of your PC or if you use multiple PC, it may be a good idea to use "LOCALHOST" instead of your real computer name (assuming the application and the database is located on the same computer)

```
DATA SOURCE=LOCALHOST\\SQLEXPRESS;DATABASE=MEASUREMENTS;UID=sa;PWD=Password123;
```

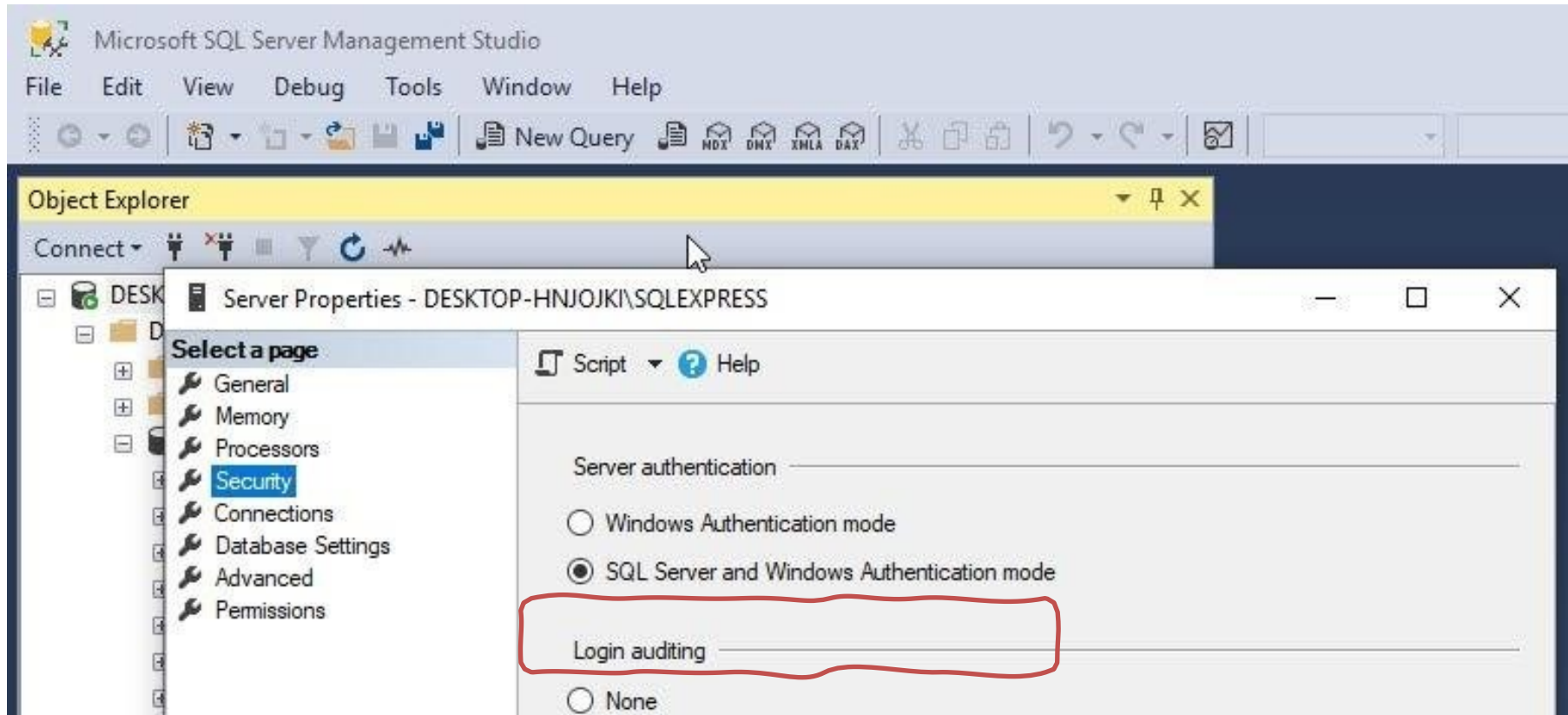
Enable SQL Server Authentication in SSMS

You can also turn on "SQL Server Authentication" in SQL Server Management Studio (SSMS) after installation of SQL Server.

To change security authentication mode, do the following steps:

1. In SQL Server Management Studio Object Explorer, right-click the server, and then click Properties.
2. On the Security page, under Server authentication, select the new server authentication mode, and then click OK.
3. In the SQL Server Management Studio dialog box, click OK to acknowledge the requirement to restart SQL Server.
4. In Object Explorer, right-click your server, and then click Restart. If SQL Server Agent is running, it must also be restarted. Or just restart your computer.

Enable SQL Server Authentication



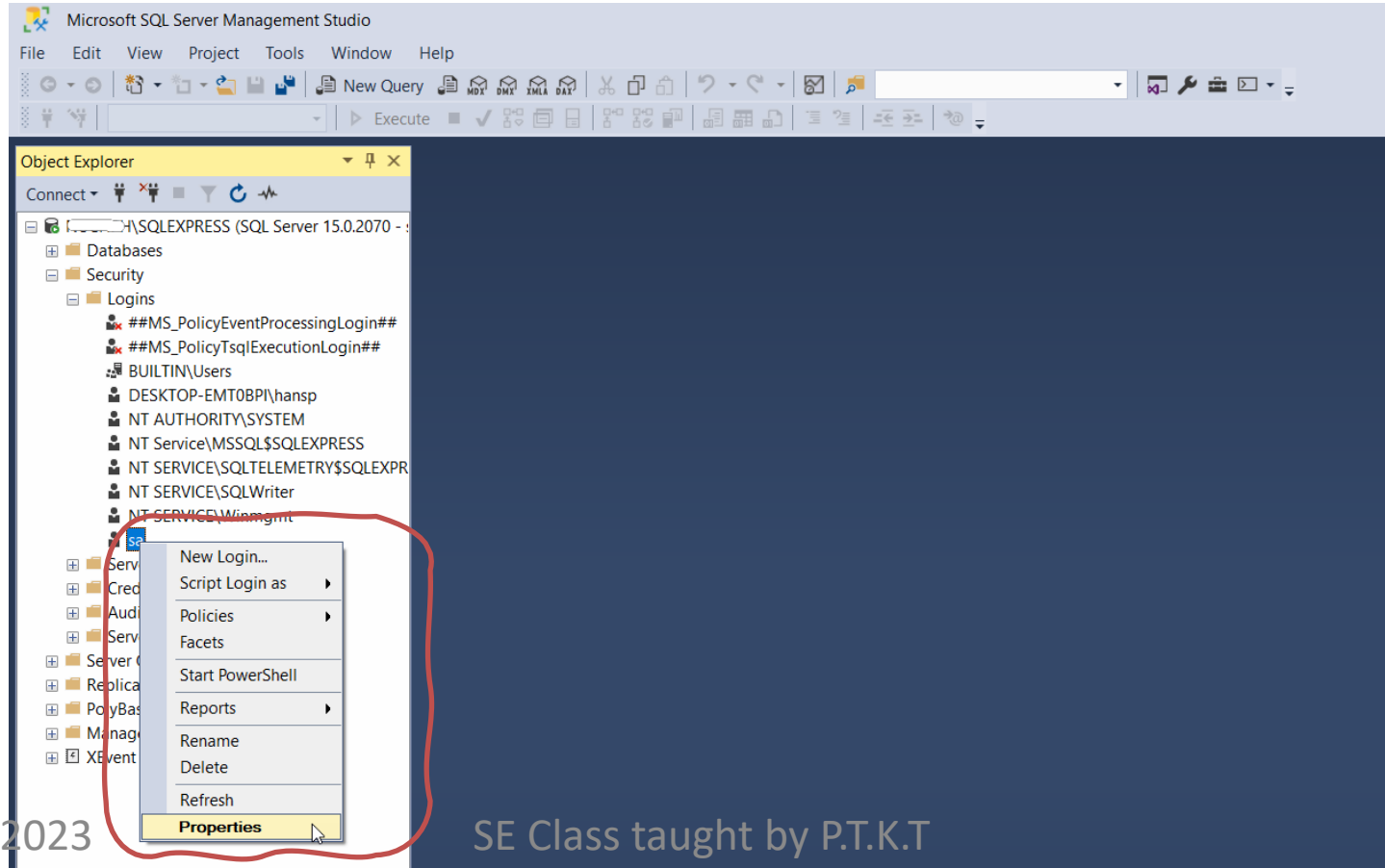
Enable sa login

Then to enable the sa login, do the following steps:

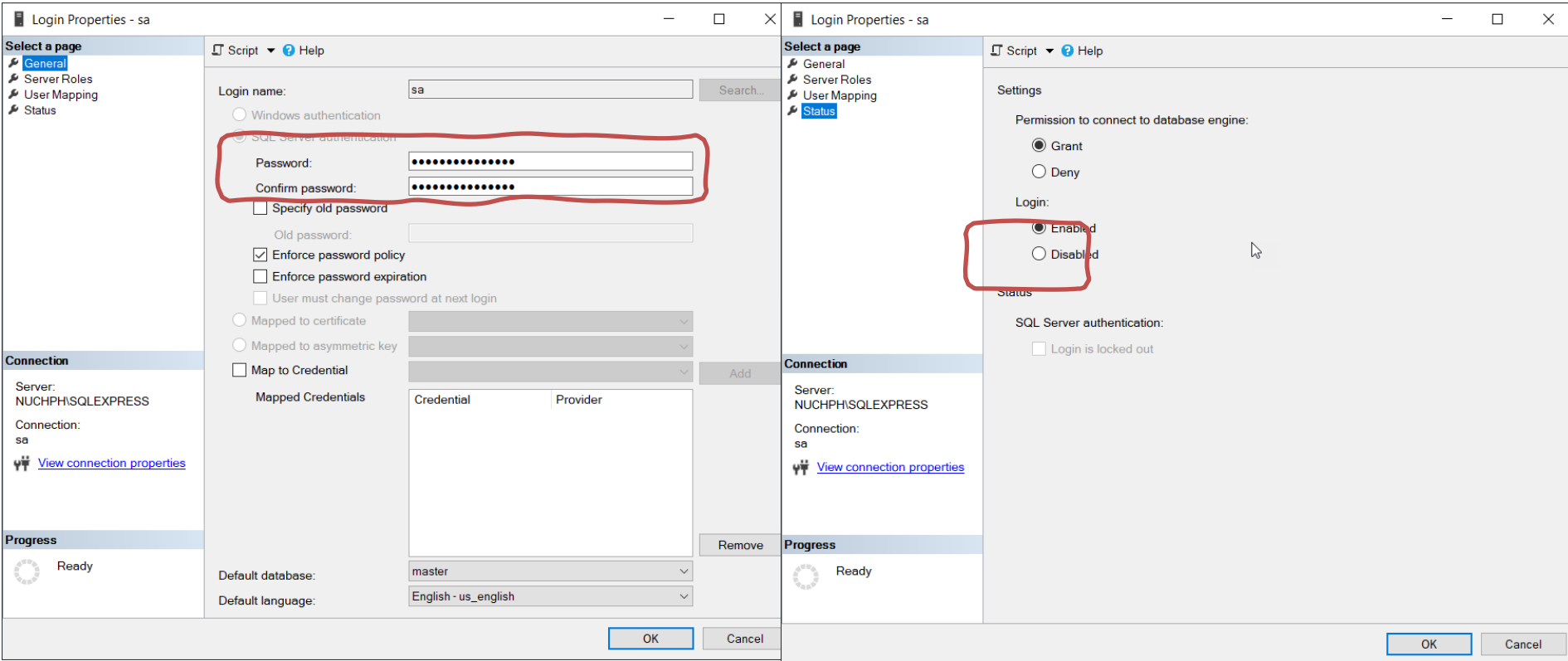
1. In Object Explorer, expand Security, expand Logins, right-click sa, and then click Properties.
2. On the General page, you might have to create and confirm a password for the login.
3. On the Status page, in the Login section, click Enabled, and then click OK.

Note! You must restart your computer afterwards (well, it is enough to restart the “Sql service..”) in order to work.

Enable sa login



Enable sa login



Create Logins in SQL Server

- “sa” is a built-in Login in SQL Server
- You can also create your own SQL Server Logins
- Normally you should do that rather than using the “sa” login
- “sa” have access to “everything” and in context of Data Security that is unfortunate.
- In general, you should make your own Logins that have access to only what's strictly necessary

Create Logins in SQL Server

In order to create a new Login, goto «Security» and right-click on «Logins» and select «New Login...»

The screenshot illustrates the process of creating a new login in SQL Server. On the left, the 'Object Explorer' pane shows the 'Security' folder expanded, with 'Logins' selected. A red circle highlights the 'Logins' folder, and a red arrow points to the 'New Login...' option in the context menu. The 'Login - New' dialog box is open on the right, showing the 'General' tab. The 'Login name' field is empty. The 'Authentication' section has 'Windows authentication' selected. The 'Password' section has 'Enforce password policy', 'Enforce password expiration', and 'User must change password at next login' checked. The 'Mapped to certificate' and 'Mapped to asymmetric key' options are also visible. The 'Default database' is set to 'master' and the 'Default language' is set to '<default>'. The 'Progress' bar at the bottom indicates 'Ready'.

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SE Class taught by P.T.K.T

Create Logins in SQL Server

Login - New

Select a page: General, Server Roles, User Mapping, Securables, Status

Script ? Help

Login name: AppLogin

☐ Windows authentication

☒ SQL Server authentication

Password:

Confirm password:

☐ Specify old password

Old password:

☒ Enforce password policy

☐ Enforce password expiration

☐ User must change password at next login

☐ Mapped to certificate

☐ Mapped to asymmetric key

☐ Map to Credential

Mapped Credentials: Credential Provider

Connection: Server: NUCHPH\SQLEXPRESS, Connection: sa, View connection properties

Progress: Ready

Users mapped to this login:

Map	Database	User	Default Schema
☒	BOOKS	AppLogin	
☐	CHART		
☐	LIBRARY		
☐	master		
☐	model		
☐	msdb		
☐	tempdb		

☐ Guest account enabled for: BOOKS

Database role membership for: BOOKS

- ☐ db_accessadmin
- ☐ db_backupoperator
- ☒ db_datareader
- ☒ db_datawriter
- ☐ db_ddladmin
- ☐ db_denydatareader
- ☐ db_denydatawriter
- ☐ db_owner
- ☐ db_securityadmin
- ☒ public

OK Cancel

You can specify which Databases that the Login should get access to and what he can do with that Database ("Write", "Read", etc.)

Windows Authentication

Windows Authentication

Using "Windows Authentication" the Connection String looks like this:

```
DATA SOURCE=DELLPCWORK\\SQLEXPRESS;DATABASE=MEASUREMENTS;Integrated  
=True;
```

Localhost:

If you don't know the name of your PC or if you use multiple PC, it may be a good idea to use "LOCALHOST" instead of your real computer name (assuming the application and the database is located on the same computer).

```
DATA SOURCE=LOCALHOST\\SQLEXPRESS;DATABASE=MEASUREMENTS;Integrated Security=True;
```

ASP.NET Core

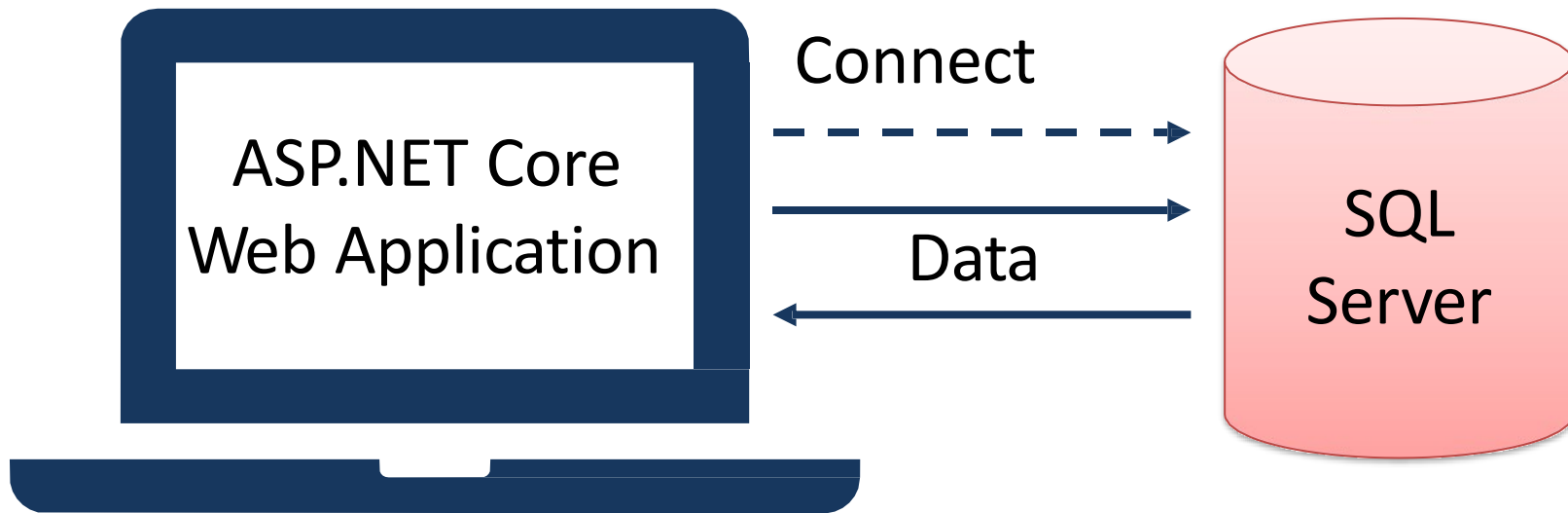
Introduction

- **appSettings.json** is a configuration file used in ASP.NET Core Web Applications
- It is typically used to store the Connection String to the Database
- But it can be used to store lots of other settings that you need to use in your application

Connection String in appSettings.json

Connection String

Connection String



ConnectionString": "DATA SOURCE=xxx; DATABASE=xxx; UID=xxx; PWD=xxx

appSettings.json

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft": "Warning",
      "Microsoft.Hosting.Lifetime": "Information"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "ConnectionString": "DATA SOURCE=xxx;UID=xxx;PWD=xxx;DATABASE=xxx"
  }
}
```

Startup.cs

We need to add something to the “**Startup.cs**” file:

```
public void ConfigureServices (IServiceCollection services)
{
    services.AddRazorPages ();

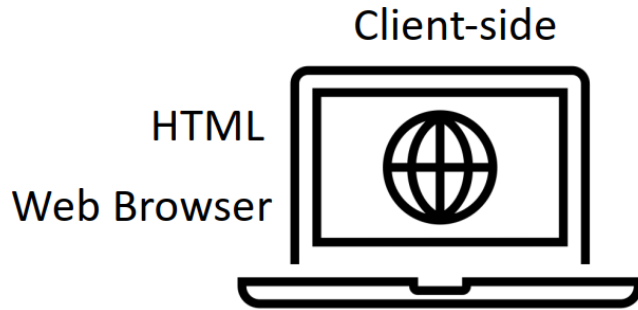
    services.AddSingleton<IConfiguration>(Configuration) ;
}
```

We have added:

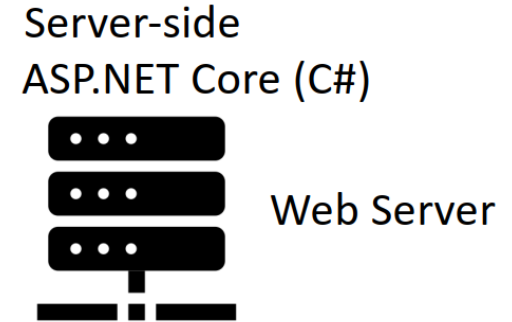
```
services.AddSingleton<IConfiguration>(Configuration);
```

Resources

- ASP.NET Core fundamentals overview : [ASP.NET Core fundamentals overview | Microsoft Learn](#)
- Introduction to Razor Pages in ASP.NET Core: [Introduction to Razor Pages in ASP.NET Core | Microsoft Learn](#)
- Tutorial: Create a Razor Pages web app with ASP.NET Core: [Tutorial: Create a Razor Pages web app with ASP.NET Core | Microsoft Learn](#)
- Razor syntax reference for ASP.NET Core: [Razor syntax reference for ASP.NET Core | Microsoft Learn](#)



Data sent between
Server and Client



localhost:7032

Hello World.

Welcome

Please enter your information:

Name:

EEmail:

Home page - HelloWorld

localhost:7032

Welcome

Please enter your information:

Name:

EEmail:

User Information: Trung Pham - phamthaikytrung@tdtu.edu.vn